DANMARKS TEKNISKE UNIVERSITET
Institute of Electrical and Photonic Engineering

BACHELOR PROJECT PLAN

107210

Design of long-range safety control for autonomous robots

Student:	August Binner Becktor
Student ID:	234396
Supervisor:	Jens Christian Andersen, jcan@dtu.dk
Date:	5. marts 2026
Duration:	Feb 2 - Jun 8, 2026 (18 weeks)
ECTS:	20

Department of Electrical and Photonic Engineering
Technical University of Denmark

1 Project Description

English

For outdoor autonomous mobile robots, the current solution relying on a gamepad is too limited. This project aims to develop a long-range wireless remote control system with safety features for autonomous robots. The system must detect loss of communication and cut power to motors, allow manual override of velocity and turn rate, and operate for at least 8 hours on battery. Optionally, the remote should display messages from the robot such as battery status.

The project involves circuit and PCB design, as well as 3D-printed casing for the devices.

Dansk

For autonome mobile udendørs robotter er den nuværende løsning med en gamepad for begrænset. Dette projekt har til formål at udvikle et langtrækkende trådløst fjernbetjeningssystem med sikkerhedsfunktioner til autonome robotter. Systemet skal detektere tab af kommunikation og slukke for strømmen til motorerne, tillade manuel overstyring af hastighed og svinghastighed og fungere i mindst 8 timer på batteri. Fjernbetjeningen skal eventuelt vise beskeder fra robotten, såsom batteristatus.

Projektet involverer kredsløbs- og PCB-design samt 3D-printet kabinet til enhederne.

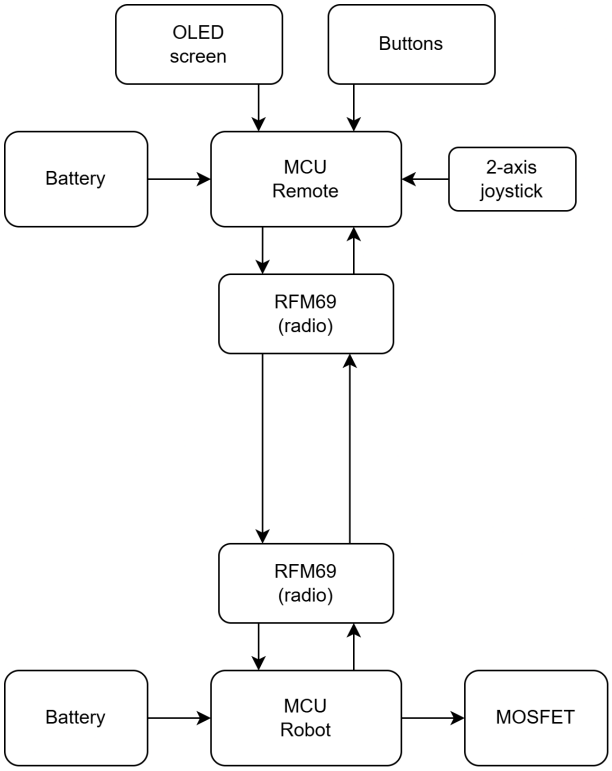
2 Project Scope

The project scope includes the following requirements:

1. **Communication Loss Detection:** System must detect when communication is lost and immediately cut power to the robot's motors.
2. **Manual Control Override:** The remote must allow the operator to take over control of the vehicle, controlling drive velocity and turn rate.
3. **Power Supply - Robot Side:** The mobile robot shall power the on-board receiver/control unit.
4. **Power Supply - Remote Side:** The hand-held remote shall be battery powered with at least 8 hours of operation on a fully charged battery.
5. **Status Display (Optional):** The remote end should be able to display messages from the robot, e.g., remaining battery power.
6. **Communication Frequency:** The system shall use the 433 MHz band for communication.
7. **Hardware Development:** The project shall include circuit and PCB design for both the remote and robot units.
8. **Enclosure Design:** 3D-printed casing shall be designed and produced for both devices.

3 Project explanation

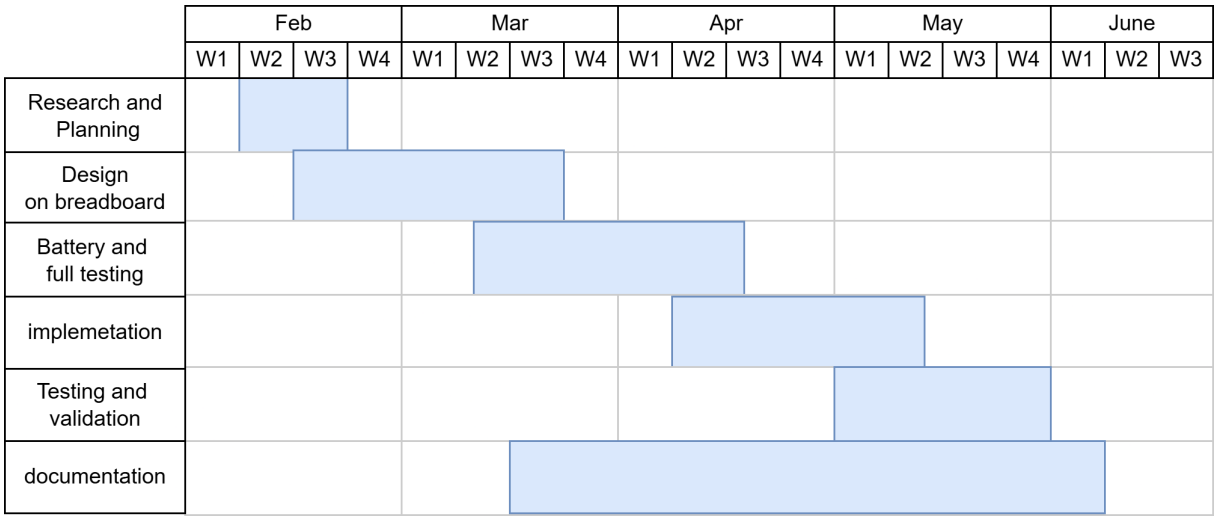
The following flowchart presents a simplified overview of how the remote control system operates. The system is divided into two main parts, each controlled by its own microcontroller (MCU). One MCU is located on the remote unit, where it handles user inputs such as the joystick and buttons and manages the display. The second MCU is located on the robot side, where it receives the transmitted signals and controls the robot's hardware components.



Figur 1: Flow chart

4 Time Schedule

The grantt chart below shows the timeline which I will be trying to follow for the bachelors project.



Figur 2: Grantt chart

5 References

Litteratur

[1] DTU - Technical University of Denmark. *Department of Electrical and Photonic Engineering*. Available at: `https://www.dtu.dk/english/education/undergraduate/undergraduate-programmes-in-danish/bsc-eng-programmes/electrical-engineering`